FINAL THESIS: DATA & BUSINESS ANALYTICS

How can an integrated analytics platform that combines customer segmentation and predictive modelling of post-purchase behaviour support profitability-oriented decision making in e-commerce retail?

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Executive Summary / Abstract

Retail and e-commerce have experienced a big increase in digital transactions, making their sector one of the most dependent on data, where businesses have to use transactional data, and consumer data to increase productivity. However, fragmentation, absence of transparency, restricted accessibility, are common problems in analytical systems used by those companies, especially the small and mid caps. Most of the systems use predictive modeling, profitability analysis and client segmentation as independent processes, which reduces their utility in the decision making processes.

Through the creation of an integrated analytical system that combines recommendation algorithms, predictive modelling of demand and client segmentation, in one unique, transparent, working pipeline, this study addresses the deficiency mentioned above. The main objective is to address the principal research question: “How can an integrated analytics platform support profitability-oriented decision making in e-commerce retail?"

The proposed solution is an integral end-to-end framework that automatically processes sets of retailers data. It uses PCA and K-means clustering to create significant segments of clients. It then applies the predictive model LightGBM to estimate reactions of demand to different price strategies and uses a recommendation system based in ALS to provide personalized recommendations to each category of customers. Additionally, the system integrates an online interactive application that allows users to modify the models parameters, add different datasets, and obtains useful results as personalised recommendations for each client and the best discount that could maximise their revenue.

What distinguishes this study from the rest is the integration and generalization of various analytical components in a unified framework. In contrast to conventional methods, the architecture proposed links explicitly the model's results with key KPIs oriented to the business, which allow users not only understand the projections but also the financial implications of every action. Furthermore, the practical usefulness of the framework increases due to the independence of the architecture of its overall data, that adapts automatically to different data types and structures through the inference schemas.

The results show that the integration of unsupervised and supervised learning methods in a single process can create valuable information for customer segmentation, pricing strategies and recommendations. The system is a useful tool that connects advanced analytics with the decision making processes from a management point of view. From an analytical standpoint, it shows how machine learning techniques can be used to address real world problems from the retail sector.

Overall, this study improves the analytical frame of the retail sector as it presents a clear, accessible and generalizable framework, which centers the decision making processes in the advanced analytics spectrum. It can make unprocessed data in strategic information, which will help give solution to day to day problems,

Literature Review

Background and motivation

The retail and e-commerce sectors are becoming more and more dependent on data due to the increase of online transactions. These online transactions create enormous amounts of data about client information, customer behavior, purchase behavior and post purchase outcomes. The market analysis shows that most retailers are moving from the decision making processes based on intuition to strategies that are based on actual data, with the objective to maintain their competitiveness in an environment which is characterised by low profit margins and high client expectations. According to McKinsey & Company (2023), advanced analytics and AI, help retailers transform operational data into useful information which allows them to offer personalized customer experiences, as well as the best pricing strategies and more efficient logistic decisions.

In the same way, Deloitte (2024) affirms that retailers consider AI and advanced analytics, key factors for productivity, engagement and loyalty of clients, even though we live in a very uncertain macroeconomic environment. Analyzing this perspective confirms that developing the capacity to obtain strategic and predictive information, form customer and transactional data is fundamental for modern retail organizations.

This need is further strengthened by the rapid development of advanced analytics for the retail market. According to market intelligence reports, the demand for customer intelligence, a unified channel of analysis and optimization instruments to boost profitability, is contributing to the global sustainable growth of analytical solutions for the retail sector

(Mordor Intelligence, 2024). Despite this expansion, many retailers still present problems with inconsistent systems and a deficient integration between the managerial decision making and the analytical results..

Customer Segmentation in Retail and E-commerce

Customer segmentation has been consolidated as a key element in the marketing and retail strategy. While data driven segmentation is more and more applied to transactional and behavioral data, traditional segmentation focuses mainly on demographic variables. Research using K-Means clustering mixed up with recency-frequency-monetary (RFM) demonstrates that customers can be divided in discrete groups with significant variations in their value contribution and purchase behaviour . (Hosseini et al., 2022).

By evaluation of cluster quality using Principal Component Analysis (PCA) and Silhouette Coefficient, traditional research confirms the efficiency of K-Means segmentation in online purchasing, demonstrating that these models may enable targeted marketing and relationship-building strategies (Liu et al., 2021). The improved personalization and the profit optimization in the e-commerce sector has also been linked to the browsing based segmentation, product preference and purchase patterns (Putra y Wadisman, 2018).

Predictive Modeling of Post-Purchase Behavior

There is an additional line of investigation that could be done that is centered in the prediction of post purchase impacts, as for example product returns, client insatisfaction and churn risks could be. A research on e-commerce return estimation confirms that supervised machine learning models, like logistic regression, random forests, and gradient boosting, can efficiently categorize orders based on return risk when taking into account discount information, product attributes, and customer behavior (Bose & Mahapatra, 2020).

According to contrasting studies, ensemble models frequently perform better than simpler classifiers by capturing non-linear correlations between explanatory factors and post-purchase outcomes, (Duong et al., 2022). In the same line, a related research on customer churn prediction in e-commerce, tree-based models could identify complex behavioral patterns linked to client churn, according to related research on customer churn prediction in e-commerce (Sunarya et al., 2021).

Existing Software Solutions and Competitive Landscape

Nowadays, a lot of retailers use commercial analytics systems instead of custom analytical tools. With the goal of immediately integrating analytics into retail decision processes, enterprise solutions like Oracle Retail Analytics offer AI-driven capabilities for pricing, inventory optimization, and forecasting (Oracle, 2023). Salesforce CRM Analytics (previously Einstein Analytics) enables customers to examine patterns and obtain AI-based suggestions by incorporating predictive insights into customer relationship management procedures, (Salesforce, 2023)

Other tools, like Adobe Analytics and Woopra, offer segmentation and group analysis features to maximize engagement and retention, with a primary focus on customer journey analysis and digital activity monitoring (Adobe, 2023; Woopra, 2023). Despite the strength of these technologies, comparable industry studies point out a number of persistent drawbacks (Retail Exec, 2024).

- First, a lot of platforms handle profitability analysis, prediction, and segmentation as different modules that need to be manually integrated. This increases complexity and reduces the practical usefulness of the analysis.
- Second, advanced analytical solutions for corporations are most of the times expensive and complicated, which makes them inaccessible to small and medium-sized e-commerce businesses. Long deployment times, specialized technical knowledge, and big financial investment are usually needed to develop the analytical systems. Despite having access to pertinent customer and transaction data, small and mid-sized businesses are unable to fully take advantage of the benefits of advanced analytics.

- Third, a lot of the mentioned analytics platforms function as "black box" systems, which provide very little insight into the underlying assumptions, models, and data transformations. These systems usually work well at providing predictions or suggestions, however their lack of interpretability damages confidence and makes it difficult for managers to understand the reasons behind results.

Porter five forces

Business analytics in the retail sector is a very dynamic and competitive industry. BIg technological companies like Oracle, Salesforce and Adobe, offer integrated analytical solutions for retail operations ,and they are the ones that dominate the industry. As a consequence, there is a big rivalry among competitors, especially in the corporate sector. However, the accessible and integrated solutions targeted to small and mid caps of online commerce lack the accessibility to these technologies as these platforms are very complex and expensive.

There is a moderate risk of new entrants. Even though cloud technology and open source tools reduce the entry barriers, the new competitors that look for a generalizable adoption face many obstacles due to the established ecosystems and strong brand identification.

Buyers have a big negotiation power, as merchants have the ability to choose between internal development, open source coding technologies or different suppliers. As a result, there exists a bigger demand for analytical suppliers that provide definite commercial value, especially when it comes to profitability and the decision making proces

Given that analytical solutions depend on software frameworks and cloud infrastructures, suppliers have a moderate amount of negotiation power. However, supplier dominance is limited because of the existence of several technology suppliers.

Lastly, the risk of substitution is moderate to high. Spreadshifts, business intelligence dashboards and internal analytical teams, are available options for the retailers. Innovative

solutions have to offer exhaustive predictive information, instead of just descriptive reports, to be able to differentiate from competition

In general, there is a high competition in the industry, but there are also opportunities for unique solutions that prioritise integration, transparency and commercial impact.

Macro-Environmental Analysis: PESTLE Framework

The retail industry is influenced by diverse external macro-environemal factors.

Data protection regulations, like GDPR, have a big impact in the way consumer data is gathered and analyzed from a political and legal perspective. The analytical platforms have to guarantee the ethical and privacy rules on data are followed.

Retailers face a bigger pressure to maximize their profitability due to economical factors such as low margins, inflation and decreased consumer spending. All these factors increase the need for solutions based on data that improve consumer retention and operational efficiency.

Additionally, retailers need to better understand consumer behavior, due to social trends such as increase in demand for customized experience and online shopping. These expectations are immediately satisfied with analytical systems that facilitate segmentation and predictive modelling.

Technologically speaking, advanced analytics is now more accessible thanks to rapid advances in cloud computation, artificial intelligence and machine learning. The necessity to compete increases with the advance and modernization of technology, because there are constantly new tools being created.

Finally, because of the environmental concerns about sustainability and efficient logistics, retailers are encouraged to improve supply chains and reduce waste. This optimizes even more the use of analytics in the operational decision making process.

These environmental factors reinforce the importance of integrated analytical strategic solutions for the retailers sector.

Gap in the Prior Art

There is a clear gap in current solutions when it comes to linking academic research with current industry practices. While industry reports confirm the high demand for retail analytics tools and the literature shows that clustering models and supervised machine learning are useful for analyzing customer behavior and post-purchase outcomes, current solutions rarely integrate these elements into a single, transparent decision-support system.

In particular, there aren't many platforms that work simultaneously:

- Customer segmentation using actual transactional and behavioral data.

Many systems of the retail sectors collect complete transactional and behavioural data, however the collected information is frequently used mainly to cluster based on rules or for the creation of descriptive reports. Rarely, the actual platforms use segmentation techniques based on data that use multidimensional points of view to dynamically capture differences in the behaviour of each client.

- Predicting loyalty-related or post-purchase outcomes.

Some platforms nowadays already have tools for predictive analysis, but they usually focus on operational indicators as demand or sales forecasting. the predictive models that focus on post purchase behavior, the long term loyalty or the satisfaction of clients are less common, and generally, they are not taken into account into the client clustering techniques.

- clear relationship between analytical results and profitability-focused KPIs.

In commercial solutions, model outputs are usually explained as abstract scores or probabilities that show no relationship with financial indicators as revenues contributors, impact of margins or the customer lifetime value could be. This makes it harder for retailers to analyse the economical impact of analytical information and to organise tasks based on predictive profitability.

- Accessibility and interpretability for decision-making in management.

Data analytics systems for the retail sector typically operate as proprietary “black box” solutions that provide little or no insight into the models, assumptions, and data transformations that underpin their results. As a result, customers may receive predictions or suggestions, such as segment labels, customer scores, or pricing recommendations, without knowing what factors influenced the outcome, how the model was trained, or why one option

is better than another. Frequently, the information is presented though complex dashboards that require technical acknowledgement for its appropriate interpretation. This could break the trust in the system and difficult the decision making process by complicating the use of secure and practical information as technical knowledge must be used. .

These limitations highlight an important vulnerability in the analytical systems of the retail sector that are used at the moment. It is necessary to address this deficiency through the creation of an integrated analytical platform, adapted to the customer data from the e-commerce industry, that is able to combine profitability analysis, predictive modelling and segmentation. All of this unified in a single, transparent pipeline oriented to the decision making process.

Research Question and Expected impact

This project addresses the following research topic in light of the stated gap:

- How can an integrated analytics platform that combines customer segmentation and predictive modelling of post-purchase behaviour support profitability-oriented decision making in e-commerce retail?

This work is anticipated to have both management and analytical effects. Analytically, it shows how realistic e-commerce data can be used to integrate supervised and unsupervised

machine learning approaches into a single pipeline. In terms of management, it offers practical insights that help with discount optimization, client priority, and retention tactics.

Stakeholders and Relevance

Many stakeholder groups can benefit from the project's outcomes. Deeper research into finding the differences in customers and profitability drivers will be beneficial for managers in retail and e-commerce. The project can be used as a guide for data analytics teams from many companies by creating comprehensible, end-to-end analytical pipelines. Lastly, scalable, transparent analytics solutions that help directly in decision-making can be beneficial for small and medium-sized companies, who are occasionally overlooked by major business platforms.

Report 1 — Methodology

Research Design and Objective

This thesis implements an integrated analytical system of data for the retail sector that combines, customer segmentation (unsupervised learning), (b) predictive modeling for demand response and profitability-oriented evaluation (supervised learning), and (c) an implicit-feedback recommender system (matrix factorization). The system is designed as a transparent and integrated work flow that links the analytical results with practical solutions (for example, segment discount policies and personalizes recommendations of products). This

design addresses directly the gap identified in the literature review explained above: there are many existing solutions that deal with segmentation, predictive modelling, and profitability analysis in separate modules, and frequently, they work as “black boxes”, which limits their interpretability and adoption from a commercial point of view.

The main research question, stated in the literature review is:

How can an integrated analytics platform that combines customer segmentation and predictive modeling support profitability-oriented decision making in e-commerce retail?

In response, the implemented process produces: (1) interpretable segmentation, (2) recommendation of discounts that optimize revenue indicators and (3) recommendation of categories based on observed interactions between customers and categories.

Software Architecture Overview

The implementación is divided into two main code modules:

- retail_models.py (Analytical Core): containing all the logic of the models, it includes the schema inferences, feature engineering, segmentation training, discount optimization, and ALS recommendations.
- app_retail.py (Streamlit Application Layer): integrating the analytical core in an interactive web interface, with caching and state management to make the experimentation easier and improve usability.

By separating the two modules instead of creating a single one be obtain generability, maintenance and clarity. (the analytical logic is independent from the logical user interface)

Data Inputs and Accepted Schema

Minimum Required Fields

The pipeline is designed to work with different CSV datasets as long as it can identify at least:

- Customer identifier (e.g., customer_id, user_id, client_id)
- Product grouping (e.g., product_category, category, product_type)

If either is missing, the system raises a validation error (to prevent misuse).

Optional Fields (Used When Available)

The accepted schema (implemented as a data class) includes fields commonly found in retail/e-commerce datasets:

- Transactions: order_id, date, unit_price, quantity, total_amount
- Discounts: discount_amount, discount_percent
- Customer descriptors: city, gender, age, returning
- Behavioral/digital: device_type, session_minutes, pages_viewed
- Post-purchase proxies: delivery_days, rating, payment_method

This structure aligns with the investigation main focus, which values the combination of transactional and conductional dimensions for segmentation and prediction tasks.

Step 1 — Schema Auto-Detection and Column Harmonization

A key code methodological contribution is the generalization between datasets through schema interfaces:

1. Column name normalization:

A normalisation function, standardises the names (transforming low cap and eliminating non-aphanumeric characters). This allows an accurate mapping between specific column names of each dataset and that accepted schema format.

2. Candidate matching strategy:

For each field (with the accepted format) (e.g., customer_id, product_category, total_amount), the code checks:

- First: exact matching between common variables (e.g., Customer_ID, cust_id)
- Second: Partial matching based in key words (e.g., “customer”, “client”, “user”)

3. Output mapping transparency:

The resulting mapping (original → accepted format)) is returned and shown in the web interface, which reduces the concerns of the “black box” issue that is highlighted in the literature review.

Rationale: The datasets of retailers are heterogeneous. A scheme interface layer allows the pipeline to be portable and backs up the thesis objective of creating a unified platform instead of a different specific script for each dataset.

Step 2 — Data Cleaning and Formatting

When the columns are identified and assigned, the preparation step guarantees that the variables have a correct format and that the missing values are handled carefully.

- The variables `customer_id` and `product_category` are transformed into a text format to ensure the creation of consistent clusters. If there is no `order_id` a temporal indicator is generated using the row index and a warning message is shown. This allows the system to continue working even if the dataset is missing explicit purchasing indicators.
- Dates are transformed into a standard date formatting. If there are any missing data values or they are not able to be transformed, neutral predetermined values are used for the functions related to time and a warning is shown to inform the user.

Rationale: this prevents the execution errors, maintaining the transparency of the processes through warning messages.

Step 3 — Feature Engineering at Transaction Level

The process generates derived variables to facilitate segmentation, discount evaluation and interpretability

Price, totals, and revenue based indicators,

When ever is needed, the code adds, creates or calculates important monetary variables:

If total_amount is missing but unit_price and quantity exist:

- Total_amount = unit_price × quantity

If unit_price is missing but total_amount exists:

- Unit_price = total_amount / quantity

with safeguards against division by zero and median fallback.

Then it creates:

order_prediscount (pre-discount revenue indicator):

- Order_prediscount = unit_price × quantity

net_revenue (post-discount revenue indicator):

- net_revenue=order_prediscount×(1−discount_percent)

value_lost_discount (discount cost indicator):

- value_lost_discount=order_prediscount×discount_percent

Rationale: the literature review emphasises the need to connect the analytical results with the KPI indicators that focus on profitability. These variables add an interpretable evaluation layer, linked to revenues even when we don't have the real margins.

Temporal features

If date is available:

- dow = day of week
- month = calendar month
- is_weekend = indicator for Saturday/Sunday

Rationale: the demand and purchasing behavior usually change depending on the day of the week and the season. These variables are useful for segmentation purposes as well as for the elaboration of predictive models.

Engagement features from digital behavior

If session and browsing fields exist (or are set to 0 otherwise):

- Engagement_score: $0.6 \cdot \log(1+\text{session_minutes})+0.4 \cdot \log(1+\text{pages_viewed})$
- high_engagement_flag: 1 if engagement is above the median.
- pages_per_minute: browsing intensity (pages divided by session minutes; safe handling for 0).

Rationale: These variables are created to change the browsing activity into significant indicators of the client interactions. The engagement_score combines the session duration and the visited pages through a logarithmic transformation to reduce the influence of the

extreme values, which are common in data form digital interaction. Adding 1 before applying the logarithm guarantees that the cases where values are 0, are processed correctly. It is given a bigger weight to the duration of the session, given that the time spent in the platform reflects a more in depth interaction than just a simple browsing among different pages. The high_engagement_flag identifies the clients whose level of interaction is higher than the median, using the median instead of the mean to avoid distorsions caused by unusual extremely high levels of activities.The pages_per_minute variable measures the brown intensity when registering the user activity in the specific content, including security measures to avoid the division by 0. As a whole, this calculation transforms digital interaction data in stable and interpretable indicators of interactions that could be combined with transactional variables for clustering and prediction.

Delivery and satisfaction risk indicators

If available:

- delivery_delay_index: Measures the difference between delivery time and the median delivery time
- fast_delivery_flag: Indicates if the order was delivered between the 25% fastest deliveries.
- low_rating_risk: Identifies the orders the the lowest qualifications from clients. (rating ≤ 2).

Rationale: These variables reflect the client experiences post-purchase. The delivery time and satisfaction qualifications are important because they can infñluence in whether a client will purchase again or not. By including these indicators, the model considers not only

the purchasing behaviour, but also the post purchase experience quality, which is linked with the client loyalty of and repeat buying.

Discount construction (real vs synthetic)

Two cases are supported:

If discount_amount exists

- discount_percent= discount_amount/ (unit_price×quantity)- clipped to a safe interval (upper bounded at 0.95).

If discount is missing

The pipeline generates a variable of an estimated % discount through a probabilistic method which is influenced by different factors.

- category propensity (e.g., fashion/home more promotion-heavy)
- price level
- weekend/campaign months
- engagement intensity

And then assigns discount values from predefined realistic discount ranges.

It also sets discount_is_synthetic = 1 when synthetic generation is used (otherwise 0).

Rationale: Many datasets don't include explicit information about discounts. Instead of stopping the analysis the code generates a transparent indicator of synthetic generation, to explicitly avoid wrong interpretations.

Step 4 — Customer-Level Feature Construction

Customer segmentation is computed from aggregated customer features:

- orders (number of orders)
- total_spend (sum of net_revenue)
- avg_basket (mean of net_revenue)
- total_qty (sum of quantity)
- avg_discount_percent, avg_engagement, avg_delivery_days, avg_rating
- recency_days: days since last purchase
- tenure_days: time between first and last purchase
- orders_per_month: normalized activity frequency
- category_diversity: number of unique categories purchased
- top_category_share: maximum share of purchases concentrated in one category
- log-transforms: log_total_spend, log_avg_basket, etc. using log1p

Aggregations convert transactional data to a customer-centric representation suitable for segmentation, consistent with segmentation research using behavioral/value variables (including RFM-like logic). Log transforms reduce skew in heavy-tailed spending distributions and improve stability of distance-based clustering.

Step 5 — Segmentation Model

Preprocessing pipeline

Before clustering, customer features are preprocessed using:

- Median imputation for numeric variables
- Robust scaling (less sensitive to outliers than standard scaling)
- Most-frequent imputation + One-Hot Encoding for categorical variables handle_unknown="ignore" to safely accept new categories at inference time

Retail data commonly contains outliers (e.g., high spenders). Robust scaling improves stability for distance-based methods.

Dimensionality reduction: PCA

Customer features can become high-dimensional after one-hot encoding. The pipeline applies Principal Component Analysis (PCA) to keep a user-selected proportion of variance (e.g., 0.90).

PCA reduces noise and multicollinearity and improves the geometry for K-Means clustering. PCA was utilized to decrease dimensionality and enhance clustering efficiency by eliminating noise and multicollinearity from high-dimensional data . For distance-based algorithms like K-Means, this enhances the data's geometric structure.

Since they are primarily intended for visualization rather than stable clustering, other techniques like t-SNE or UMAP were not employed since they decrease interpretability. PCA is more appropriate for this application since it offers a predictable, effective, and comprehensible transformation.

Clustering: K-Means with model selection

The segmentation approach trains K-Means across a range of K values (k_min to k_max) and evaluates each candidate using:

- Silhouette score (cluster separation/compactness)
- Minimum cluster proportion penalty
- composite=silhouette×penalty
- where the penalty down-weights solutions with very small clusters (below a configurable threshold).

K means was selected because of its solid performance, scalability, and ease of use, for consumer segmentation tasks using transactional and behavioral data. It enables the organization of clients in a clear and intelligible way for management decision-making.

The different cluster densities present in retail data are difficult for DBSCAN to handle, while other approaches like hierarchical clustering are less scalable and more computationally costly. K-Means offers a a better trade-off between interpretability and efficiency when paired with PCA and silhouette-based validation (Liu et al., 2021).

The added minimum-size penalty is a realistic administrative limitation because extremely small segments might be unstable and less actionable

Selection of K and cluster quality metrics (k, silhouette, min_segment_prop, balance_penalty)

The process determines the optimal number of clusters (k) through the grid-search inside a predetermined range defined by the userK ∈ [k_min, k_max]. For each possible value of K, a K-Means algorithm is applied to the space features transforming them using PCA. The results are then evaluated using three indicators which are shown in the web interface: silhouette score, the minimum segment proportion (min_segment_prop), and a size-based adjustment factor (balance_penalty). The final value of k is chosen depending on a composite score that balances the statistical separation with practical usability, avoiding too small segments to be operationally significant.

- The silhouette score measures how well a client adjusts to its assigned cluster in comparison to other clusters. It is calculated by assigning distances in the reduced PCA space, comparing the average distance of a client to other clients in the same clusters with the distance of the closest alternative cluster. The silhouette value for each observation i is defined as s(i) = (b(i) − a(i)) / max(a(i), b(i)), where b(i) is the minimal average distance to points in the closest alternative cluster and a(i) is the average distance to points in the same cluster. The total silhouette score is calculated by taking the average of s(i) for each client. Well-separated clusters are indicated by values closer to 1, overlapping clusters are indicated by values close to 0, and systematic misassignment is suggested by negative values.

- `min_segment_prop` is the minimum segment proportion. The model calculates the percentage of clients in the smaller cluster through the formula, `min_segment_prop = min_k (n_k / N)`, where n_k is the size of the cluster k and N is the

total number of clients. This step is necessary because very small clusters are usually unstable and very difficult to operationalize (for example, due to the insufficient observations for the forecasting of discounts for each segment). To check if the clustering generates segments that are smaller to “normal” ones, the value is shown in the interface.

- `balance_penalty` is the penalization for unbalanceness. The model imposes a penalization in relation with a predetermined minimum accepted percentage of acceptable number of clusters (`min_prop`), to avoid solutions for clusters with a smaller size than a practical one. The penalization is calculated through the formula `balance_penalty = min(1.0, min_segment_prop / min_prop)`. The penalization is equivalent to 1 (no penalization) when `min_segment_prop` ≥ `min_prop`. The penalization drops below 1 and the possible solution is down-weighted when min_segment_prop < min_prop. The predetermined value of min_prop in the implementation is given as 0,08 (8%).

- Composite score and K selection. For every candidate of K, the process calculates composite_score = silhouette × balance_penalty and then selects k that produces the highest composite_score. This rule guarantees that the final segmentation represents and equilibrium between statistical separation (Silhouette) and commercial usefulness (avoiding very small segments)

The literature supports K-Means segmentation in the retail sector and the use of PCA + Silhouette based evaluation to evaluate the quality of clusters. The additional penalization

for the minimum size of each cluster is a practical managerial constraint, as extremely small clusters are unstable and not successful for commercial purposes.

Optional balanced assignment (post-processing)

If enabled, the pipeline reassigns customers to cluster centers while enforcing near-equal cluster parameters. The method:

- computes distances to each cluster center,
- prioritizes “high-confidence” assignments (largest distance margin between best and second-best cluster),
- fills established clusters to avoid tiny segments.

This produces similarly sized segments for interpretability and for downstream discount estimation stability (small segments can fail minimum sample thresholds).

Additionally, making a balanced number of clusters helps prevent the overfitting in the prior steps to the predictive modelling. When clusters are extremely small, the trained predictive models at a segment level can adapt too much to the limited observations inside that specific small segment, capturing noise instead of general behavioral patterns. By guaranteeing that each segment has a sufficient number of clients, the balanced assignment reduces variance in the model estimation process, improving the trustworthiness of the statistic and makes optimizational results more generalizable. In this way, the balance procedure contributes not only to the interpretability, but also to the robustness of the model.

Step 6 — Discount Optimization by Segment

The discount policy module estimates the best discount among a candidate set (e.g., 0%, 5%, 10%, …) for each segment.

Target definition

The demand model predicts:

- y=log(1+quantity)

This stabilizes variance and limits the influence of extreme quantities.

Model: LightGBM Regressor (Gradient Boosted Trees)

For each segment, the code trains a regression model LightGBM with a mixed-feature preporecessing proces (numeric scaling + categorical one-hot encoding). Inputs include:

- product_category, unit_price, discount_percent
- time features (month, is_weekend)
- engagement and post-purchase features (engagement_score, delivery_days, rating)
- customer indicators (returning) and optional demographics/device/payment features in the case they are available

Gradient boosting is commonly used for the analysis of retail data because it captures the effects and nonlinear interactions in tabular data and works well with heterogeneous features. Ensemble methods usually outperform more simple models when it comes to the prediction of behavioral results.

For tabular retail data, where the correlation between variables as price, discounts and consumer behavior are usually non linear and imply complex interactions, LightGBm was chosen because of its high effectiveness. It has been demonstrated that gradient boosting models outperform simpler models in behavior predicting tasks and increase the prediction precision. (Bose y Mahapatra, 2020; Duong et al., 2022)

LightGBM captures more realistic demand patterns than linear regression since it does not require linearity. Because of its boosting process, it offers more predictive performance and efficiency than Random Forest. Neural networks were not chosen because they are not suitable for segment-level modeling due to their increased complexity, reduced interpretability, and larger data needs.

For demand estimates in this architecture, LightGBM offers the best overall balance between accuracy, scalability, and usefulness.

Revenue objective function (per candidate discount)

For each candidate discount r, the code simulates the same orders under r by overriding discount_percent = r, predicts the expected quantity $q^{\wedge}$, and computes expected revenue proxy:

- ExpectedRevenue(r)=E[unit_price×(1−r)×q^]

The recommended discount is:

- $r* = \arg\max \text{ExpectedRevenue}(r)$

Synthetic-discount fallback (“elasticity prior”)

If the dataset originally lacked discounts (synthetic discount mode), the code avoids fitting a misleading demand model (because there is no real variation to learn from). Instead, it uses a simple elasticity-like rule:

assume demand grows exponentially with discount:

- $q^(r)=q^{-}\cdot e\beta r$
- compute expected revenue with the same net price adjustment and choose the best r.

This makes sure that the application still works correctly with different datasets, at the same time that it clearly labels the method as heuristic when data don't allow causal/associational learning of discount effects.

Computational safeguards

To avoid UI freezes and unstable estimates, the implementation includes:

- min_rows_segment threshold (skip segments with insufficient observations)

- sampling cap per segment (max_rows_per_segment)
- limit on number of segments to process

Step 7 — Category Recommendation System (ALS)

The recommendation components generate recommendations of personalized categories through implicit feedback.

Interaction matrix construction

A sparse customer–category matrix is built where:

- rows = customers
- columns = product categories
- values = interaction weight (default: purchase quantity)

Model: Alternating Least Squares (ALS) for implicit feedback

The code uses the factorization matriz ALS of the Implicit library with latent configurable factors by users.

ALS is a standard focus for the implicit recommendation (visualization, purchases, frequency) because it adapts correctly to sparse matrices and creates significant representations of latent preferences.

Output interpretation

For a chosen customer, the models returns the N main recommended categories including:

- Raw score: it is an estimation of the ALS model of the strength spectre from the association between one client and one category. It represents the probability that a customer would choose or engage with the category recommended and it is generated from the interaction of the latent variables, found during the matrix factorization step. Nevertheless, there is no direct probabilistic explanation for this punctuation, knowing that it is relative.

- Normalized affinity (%) relative to the maximum score obtained: scaled version of the raw score, calculated in relation to the maximum score of the clients in the suggested categories. The preference is expressed as a percentage, where the 100% indicates the most important category. This standardisation improves interpretability and comparison of the suggestions.

This provides a user friendly output, consistent with the objective thesis of improving the decision making processes.

ALS was chosen because it is specifically designed for implicit feedback data, like purchases and interaction frequency, which are common in teh retail sector. It generates latent representations for the recommendations and it efficiently manages sparse and massive user-item matrixes.

Because of the scarcity of metadata from products, no other strategies were used, like a content based filtering. Neither were used other collaborative neuronal filtering due to the bigger computational cost and complexity. ALS offers the best equilibrium between scalability, ease of implementation and performance.

Reproducibility and Transparency Features

Random seeds (e.g., 42) which are fixed, are used across PCA, K-Means, synthetic discount generation, and sampling steps.

The UI displays:

- mapping detected (original → canonical(acepeted)),
- warnings for inferred/missing fields,
- segmentation metrics and segment profiles,
- method labels for discount computation.

These choices directly address interpretability concerns raised in the literature review about “black box” analytics platforms.

Software Environment and Libraries

The retail dataset's analytical pipeline was built using Python and a variety of libraries that were chosen for how well they would integrate into the data science workflow, how commonly they are used within machine learning applications and because they would allow for scalability, repeatability, and easy integration between the data processing, machine learning, and user interface components.

To achieve these benefits the primary libraries used for processing data and performing numerical calculations on the retail dataset were Pandas and NumPy. Pandas provided a way to manage tabular data which supported the execution of filtering, aggregating and transforming of transactional data. NumPy enabled efficient low-level numerical calculations to augment the functionality provided by Pandas to improve efficiency while working with large datasets.

Scikit-learn was utilized to build the machine-learning pipeline. It supports pre-processing steps like missing-value treatment, scaling of numerical features, and also the encoding of categorical features; additionally, it has been used for the modelling processes of K-Means and PCA. Scikit-learn provides a modular and consistent framework, which allows easy combination of pre-processing steps and modelling steps to ensure reproducibility.

The LightGBM library was used for predictive modelling. LightGBM is a gradient boosting framework that performs exceptionally well on structured tabular data and is

computationally efficient compared to other similar implementations. Its ability to model both linear and non-linear relationships between variables makes it the most suitable choice for modelling retail demand behaviour.

The recommendation system was built using the Implicit library that utilizes the ALS algorithm. This library is tailored to work with implicit feedback data (i.e., purchase interactions) and enables efficient handling of large, sparse user-item matrices. Additionally, SciPy was used to create representations of the scattered matrices, which results in fewer requirements for memory and better computational performance.

Streamlit was used for developing the user interface of this project; it provides an easy way to create interactive applications that display data. With Streamlit, users have the ability to upload their datasets, change the parameter settings of each mode, and see how their changes impact the outcome based on a sample dataset, all without writing complex front-end code. Streamlit was chosen because it is user-friendly and integrates seamlessly with Python-based analytical processes.

The overall use of all these technologies creates an integrated environment supporting modeling, data preparation, visualization, and user interaction. Together, these technologies enable efficient, scalable, and aligned solutions for providing transparent retail analytics decision support solutions.

Web Application Report

Purpose and End-to-End Workflow

The web app titled "Retail Sales Optimizer (General)" implements the following workflow:

1. Upload or load a CSV dataset
2. Auto-detect and harmonize columns (schema inference)
3. Create engineered variables (preview shown)
4. Build customer features and segment customers
5. Compute segment-level optimal discount policy
6. Train ALS recommender and generate customer-level category recommendations

Left Sidebar Controls (User Inputs)

1) Dataset

- "Upload your dataset" (File uploader): Uploads a custom dataset.
- "Use example dataset if no other CSV is uploaded" (checkbox): Loads a default example dataset if no file is uploaded.

This enables: change of dataset without the necessity of rewriting the code, because the system automatically detects columns and does a flexible cleaning. The application updates automatically the mapping, variables, segments and recommendations.

2) Segmentation (Customer Segmentation)

- “Min segments (k_min)” slider: Lower limit for the minimum number of clusters explored.
- “Max segments (k_max)” slider: Upper limit for the maximum number of clusters explored.
- “Compression level (PCA)” slider: Controls the variance retention of PCA (eg, 0,90 retains approx the 90% of variance)
- “Balance segments (similar sizes)” checkbox: Applies the balance assignation of the postprocessing step so that the segments have similar sizes

Why this matters:

Increasing K produces a granular segmentation, but it can also reduce stability. Additionally, PCA controls complexity and improves the geometry of clusters. Lastly, rebalancing reduces small clusters that are difficult to interpret and too small to estimate a proper and effective discount.

3) Optimal discount per segment

- “Discounts to test (%)” multiselect: show the available discounts tested (e.g., 0, 5, 10, 15, 20, 25).
- “Minimun rows per segment to estimate” slider: Minimum size of data required in each segment to calculate a discount policy
- Anti-freeze controls:
    - “Max rows per segment (simulation)” slider:Limits the training rows by segment (performance protection)

- “Max rows to train global demand” slider: Appears like user interface control for the protection and extension of the calculations. In the actual implementation, the discount modelling is at a segment level, so this variable works mainly as a position indicator for future extensions and limitations.
- “Calculate optimal discounts” button: It activates the complete calculation of discounts (most complex step)

It is important to highlight that the application uses the bottom (not an automatic recalculation) to avoid training the model repeatedly when the user adjusts the slider controls.

Output meaning: the resulting table shows for each segment:
Number of used orders, approximation of expected revenues with the selected discount, Methods used (segment_demand_model vs heuristic prior) and a Calculation status (“OK” or “Not calculated”)

4) Recomendaciones (ALS)

- “Nº of recommended categories" slider: Number of categories returned.
- “Recommendator quality(factors)” slider: Size of the latent dimension (more factors can improve the adjustment, but can make an overadjustment or increase the calculation)
- “Train recommendator” button: Trains ALS once (again using a button to avoid continuous retraining).

After training, the UI provides:

- a Customer_ID selector
- the customer's segment label
- the segment's recommended discount
- a "Category Recommendator" button that displays recommended categories with score and affinity.

Main Page Outputs (What the User Sees)

Section 1 — Dataset Preview

The app displays:

- dataset source (uploaded vs example),
- number of rows and columns,
- a preview table.

This gives immediate feedback that the file was loaded correctly.

Expanders — Warnings and Mapping

- "Warnigns (per missing columns/ inferences)" shows warnings when the pipeline creates missing fields (e.g., synthetic order_id, default quantity, synthetic discount).
- "Mapping detected (original column → correct format)" displays the inferred mapping dictionary.

Why it matters: This is a transparency feature that reduces black-box behavior and supports the thesis emphasis on interpretable decision support.

“Variables created “(preview)” — Engineered Variable Panel

This panel is central to your thesis explanation: it shows the engineered variables created automatically, including:

- discount_percent (+ discount_is_synthetic flag)
- order_prediscount, net_revenue, value_lost_discount
- engagement variables (engagement_score, high_engagement_flag)
- delivery and rating risk flags
- time variables (dow, month, is_weekend)

What it demonstrates: The system does not only run models; it constructs interpretable analytical variables that later feed segmentation and optimization.

Segmentation Outputs

The app shows:

- segmentation metrics (selected K and silhouette-based indicators),
- a bar chart of segment sizes,
- a segment profile table (mean spend, recency, engagement, etc.)
- Interpretation: This table explains “who is who” across segments, improving managerial usability.

Discount Policy Outputs

Once computed, the discount table shows the expected revenue proxy (per segment), selected discount and status and method used.

This provides a direct “decision artifact” (a policy) rather than only a model score.

ALS Recommendation Outputs

The recommendation table includes, recommended categories, a score, and a percentage of affinity.

This provides a second decision artifact: “what should this customer likely buy next (at category level).”

Methodological Limitations and Implementation Challenges

The proposed analytical pipeline was developed successfully, however, during the implementation phase, various methodological and practical problems were faced. These restrictions reflect the difficulty to combine a wide range of analytical elements, as well as the limitations of the data of the retail sector..

Evolution of the Demand Modeling Approach

Accurately estimating the demand response to discounts was one of the major issues the project faced. Firstly, it was expected that demand followed a predetermined exponential relationship, with the levels of discount through a simple method based on a constant price elasticity assumption. Even though this method offered a fast approximation, it turned out to be very limited and unable to capture the real complexity of customers.

As a consequence, LightGBM was incorporated. It is a predictive model that learns demand patterns directly from the datatset. This change, increased considerably the asdapatibility and accuretneess of the model, allowing it to represnrt interactions and non-limear correlations between variables. . The estimated effects of discounts should be understood as correlations rather than actual causal consequences because the final model is associational rather than causal.

Generalization Across Datasets

Making sure that the workflow was useful for as many companies as possible in the retail sector was another important obstacle faced. The version of the system was dependent on a specific format of data with predetermined variables and names of columns. This limited the use and was contradictory with the main objective which was creating an analytical tool that was reusable and adaptable.

To address this problem an inference and autodetection schema was created, which allowed the system to find relevant columns through keyword and naming patterns. Even

though this addition considerably increased the generalization ability of the system, it brought some disadvantages, as the datasets with less common names or not very clear ones could make the labelling be inaccurate.

Web Application Development Constraints

There were difficulties in the development of the user interface. For the first attempt, Gradio was used, which was very helpful for a first attempt due to its simplicity and prototyping capabilities, but it failed to satisfy the project requirements, specifically when managing complex workflows, dynamic state management and multi-step analytical procedures. It turned out be useful to test individual model components and create a basic interactive interface. Given that the analytical system requires sequential processes (segmentation, discount optimization, and recommendation), maintaining intermediate results Gradio's structure made state management and conditional execution difficult to control.

As a result, Streamlit was chosen to implement the application, adding a better compatibility with the state persistence and the interactive dashboards. Unlike Gradio, streamlit allows a better control over the flow application, facilitating the management of multi step analytical processes, such as dataset upload, segmentation, discount optimization, and recommendation within a single coherent workflow. Its function of session state enables intermediate results to be stored and reused within user interactions, which is essential to maintain consistency when parameters are modified.

Additionally, streamlit integrates naturally with the chosen libraries and allows for dynamic visualizations, tables, and conditional execution logic, making computationally

intensive models to run only when explicitly asked to. Although streamlit is not designed for large scale implementations and could bring limitation drawbacks as opposed to DJango or Flask, it offers an excellent equilibrium between usability, rapid development and perfect integration with analytical pipelines.

Data Limitations and Synthetic Feature Generation

The disponibility of discounted data presents a big limitation. Explicit discount factors are extremely important to estimate price sensitivity, but are not available in every retail dataset. To address this problem, a model that creates synthetic discounts based on contextual factors (time, category, interaction) and probabilistic assumptions was created.

This method adds an additional approximation layer, but it allows that the process works without discount data. Given that the results from synthetic discounts are based in simulated circumstances instead of real market activity, they have to be evaluated with awareness.

Segment-Level Modeling Constraints

The decision of training predictive models at a segment level brings data-level limitations, but it also improves interpretability and is consistent with managerial use cases. Some segments can have just a small amount of observations, which could reduce the precision and stability of the model predictions. When a segment is too small, the predictive model can have difficulties to identify consistent patterns, and adapt to the specific available data points. Security measures were implemented, like minimum sample thresholds and data

controls, to reduce this risks, however the reliability of results could still change depending on the segmentation structure and total amount of data in the dataset. In sampler datasets, or when clusters are very unbalanced, the performance of the model could be less consistent between models.

Future Research Directions

There are many ways in which this project could be further developed. Right now, the optimization process is based on the maximization of revenues and does not explicitly take into account costs or profit margins. As a consequence, it is true that the proposed discounts could increase the sales amounts and the total revenues, however they don't necessarily guarantee a higher profitability. A logical next step would be to introduce a Cost of Goods Sold (COGS) variable, which would allow the model to move from an optimization based on revenues to an evaluation based on real profits. By introducing this cost info in the calculator, the system would estimate the net profit outcomes and recommend discount strategies that would genuinely improve the financial performance. This would make the model more realistic and would better align with the commercial objectives of a sustainable profit increase instead of simply increasing sales.

Secondly, even though the model detects patterns in data, it does not demonstrate that discounts really modify the customer behavior. To have a better understanding of the discount effects, future work should add more sophisticated techniques, like experiments of causal models.

Also, a relevant step for future research would be an automatic actualization system implemented through an API connected to the data sources of the company that is working

for. At the moment, the process requires human intervention to load the datasets and execute the models. In the future, an architecture could be developed that allows direct connection with business data or ERP systems through APIs, so that the information updates in real time or programmed time intervals. This would allow the system to work in an automatic way, without manual intervention, generating segments, recommendation and discount strategies in a continuous way. This type of integration would increase the practical applicability of the model, and would make it something more useful for real world application, where decisions have to adapt dynamically to changes in the consumer behavior and market conditions.

Lastly, the model could be improved by making the system more scalable and more appropriate for a practical implementation. This implies to optimize the performance of bigger datasets and integrate them with streaming data.

Practical Implications and Real-World Application

The study results found that the proposed analytical process is directly applicable to the day to day decision making process in the retail sector. The model combines segmentation, prediction and recommendations in a unified workflow, which allows to develop efficient commercial strategies, as opposed to the analytic and traditional focus that offer discrete insights.

From the management standpoint, the results of the segmentation allow companies to identify consumer groups with different levels of involvement, purchasing patterns, and discount sensitivity. This makes the targeted marketing strategies possible, for example,

offering higher discounts to the segments that are more sensible to price changes, maintaining the profitability to the less sensible customers.

By offering segment-level pricing recommendations based on data, the discounts optimization component facilitates the decision making process. The findings demonstrate that different segments have different ideal discount levels, underscoring the possibility of less-than-ideal results from uniform pricing schemes. This illustrates the importance of personalised pricing strategies to increase profits.

Additionally, by highlighting the categories with a higher projected affinity, the recommendation algorithm offers valuable information about customer preferences. This could be applied to improve the user experience, boost sales perspectives and increase cross-selling and customization efforts.

Data Sources

The datasets used in the modeling of this project were provided by the online platform Kaggle, which offers publicly accessed datasets for the investigation of data science and machine learning research. Due to its accessibility, diversity and applicability to real world problems, Kaggel datasets are frequently used in academic and practical investigation.

The selected datasets include characteristics such as product, price, size and client interactions, as well as transactional and relational information about clients, which are

typical in online retail. The datasets were selected because they allow technical application of segmentation and predictive modelling and offer a realistic representation of retail operations.

The use of public data, which allows transparency and reproducibility of the research, was one of the main concerns of the project. However,extra preprocessing and feature engineering procedures were needed to adjust the data to the analytical framework because Kaggle datasets are frequently anonymised and might not contain all pertinent business factors (such cost or specific promotional data).

In general, given that the system was created to work with various datasets, instead of depending on one unique proprietary source, the inclusion of Kaggle datasets, contributes to the objective of creating a unique generalizable and reusable workflow.

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